Course Syllabus

General Information

Faculty	Information Technology		
Department	Cybersecurity Dept.		
Semester	3rd	**Academic Year**	2025/2026

Course Details

Course Title	Secure Communication Protocols بروتوكولات الاتصال الامنة				**Course No.**	A0433403
Type of Learning	☐ On- campus ■ Blended (Moodle Platform) ☐ Online (………... Platform)					
Credit Hours	3	**Theoretical**	3	**Practical**	0	
Course Level (according to JNQF)		7	**JNQF Hours***	12.15		
Pre-requisite	A0432401- Cryptography Fundamentals			**Co-requisite**		

Course Instructor Information

Instructor Name	Dr. Hashim Jarrar		
Instructor E-mail	h.jarrar@meu.edu.jo		
Course Time	9:30-11:00 (Sun, Mon)		
Office Hours	11-12(Sun, Mon)		
Office Number	N133	**Office Phone**	512
Lab Instructor Name –If assigned-	--		

Sources and References

1. Textbook

Book name, author, publisher, edition, publishing year

Computer Security: Principles and Practice, Stallings, W., & Brown, L., Pearson Education Limited, (5th ed.), 2024

2. References

1. CompTIA Security+ Guide to Network Security Fundamentals, Ciampa, M., Cengage Learning, (7th ed.), (2022).
2. Cryptography and Network Security: Principles and Practice, Stallings, W., Pearson, (8th ed.), (2019).

Teaching Methods

1. Lectures
2. Quizzes
3. Micro Credential Certificate
4. Assignments
5. Discussion & Questions
6. Exams

Course Description

This course provides a comprehensive study of secure communication protocols and their operation across the OSI model layers, with emphasis on security mechanisms at the network, transport, and application layers. Students examine the design, architecture, and security properties of widely deployed protocols including **TLS, HTTPS, HTTP/3 DNSSEC, SSH, IPsec, RADIUS, TACACS+, Kerberos, S/MIME, and PGP**. The course emphasizes cryptographic foundations, authentication mechanisms, key management, and trust models used to secure data in transit. Student analyze protocol vulnerabilities, evaluate security requirements, and design secure communication solutions for enterprise cybersecurity scenarios. Building on prior knowledge of cryptography and network fundamentals, this course prepares students for advanced security practice in infrastructure protection and incident response.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	**Assess** the architecture, operation, and security mechanisms of core secure communication protocols **(TLS, HTTPS, SSH, DNSSEC),** including their role and vulnerabilities in protecting data in transit.	PLO1		S1-P	
2	**Assess** secure communication protocols **(IPSec, VPNs, RADIUS, TACACS+, Kerberos, PGP, secure email)** in terms of security properties, deployment scenarios, and limitations using a CIA and adversarial thinking framework.	PLO6		S1-P	
3	**Design** a secure communication architecture for an enterprise scenario by selecting and integrating appropriate protocols to address identified threats and constraints.	PLO1		S1-D	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	**Introduction to Secure Communication Protocols:** OSI model security layers, TCP/IP vulnerability overview	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
2	**SSL (Legacy) and Evolution to TLS Protocols**	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
3	**TLS protocol and Handshake Mechanisms:** TLS handshake, record protocol, cipher suites, TLS 1.2, 1.3, vulnerabilities	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
4	**HTTPs and HTTP/3 Security**	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative

5	**DNS Security:** DNS vulnerabilities (spoofing, cache poisoning), DNSSEC, DNS-over-TLS (DoT), DNS-over-HTTPS (DoH)	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
6	**SSH Protocol:** SSH architecture (transport, user authentication), key exchange, port forwarding (tunneling), SFTP, SCP	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Quiz1 (7%)	Summative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
7	**VPN Fundamentals:** IPSec architecture, Transport mode vs. Tunnel mode, Security Associations (SA), SPD, SAD. Protocols AH, ESP	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
8	**IPSec Protocol:** IPSec architecture, Transport mode vs. Tunnel mode, Security Associations (SA), SPD, SAD. Protocols AH, ESP	Lecture (1)	on campus	1	Discussion & Questions	Formative
		Lecture (2)	on campus	1	Discussion & Questions	Formative
		Lecture (3)	Embedded	1	Discussion & Questions	Formative
9	**Mid Term Exam (25%)**	Lecture (1)	on campus	1	Questions	Summative
10	**RADIUS protocol:** AAA model, EAP integration, 802.1X (port-based authentication)	Lecture (1)	on campus	2	Discussion & Questions	Formative
		Lecture (2)	on campus	2	Discussion & Questions	Formative
		Lecture (3)	Embedded	2	Discussion & Questions	Formative
11	**TACACS+ protocol:** architecture, comparison with RADIUS, deployment scenarios	Lecture (1)	on campus	2	Discussion & Questions	Formative
		Lecture (2)	on campus	2	Quiz2 (8%)	Summative
		Lecture (3)	Embedded	2	Discussion & Questions	Formative
12	**Kerberos Authentication Protocol:** ticket-granting architecture, AS and TGS roles, replay attack protection	Lecture (1)	on campus	2	Discussion & Questions	Formative
		Lecture (2)	on campus	2	Discussion & Questions	Formative
		Lecture (3)	Embedded	2	Discussion & Questions	Formative
13	**Email Security Protocols:** SMTPs, IMAPs, POP3, SMTPs, MIME & S/MIME Protocol	Lecture (1)	on campus	2, 3	Discussion & Questions	Formative
		Lecture (2)	on campus	2, 3	Assignment (10%)	Summative
		Lecture (3)	Embedded	2, 3	Discussion & Questions	Formative
14	**PGP Protocol:** Integrated Protocol Architecture Design Workshop (Secure Communication Case Study)	Lecture (1)	on campus	2, 3	Discussion & Questions	Formative
		Lecture (2)	on campus	2, 3	Discussion & Questions	Formative
		Lecture (3)	Embedded	2, 3	Discussion & Questions	Formative
15	**Micro Credentials Certificate (10%)**	Lecture (1)	on campus	1,2,3	Discussion & Questions	Formative
		Lecture (2)	on campus			
		Lecture (3)	Embedded			
16	**Final Exam (40%)**	Lecture (2)	on campus	1,2,3	**Exam**	**Summative**

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Quiz 1	**7**	**Week 6**
Quiz 2	**8**	**Week 11**
Assignment	**10**	**Week 13**
Midterm Exam	**25**	**Week 8**
Project	**10**	**Week 15**
Final Exam	**40**	**Week 16**
Total	**100**	**Week 6**

Item	Policy
Quizzes	▪ No makeup quizzes. ▪ A minimum 2 quizzes are given. ▪ If five quizzes or more are given then the lowest quiz's grade is dropped.
Exams	▪ The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	▪ Makeup exam should not be given unless there is a valid excuse.
Drop Date	▪ Last day to drop the course is according to the University regulations.
Academic Honesty	▪ Cheating or copying on exam or quiz is an illegal and unethical activity. ▪ Standard MEU policy will be applied. ▪ All graded assignments must be your own work (your own words).
Attendance	▪ Excellent attendance is expected. ▪ MEU policy requires the faculty member to assign ZERO grade if a studentmisses 20% of the classes. ▪ Sign-in sheets will be circulated. ▪ If you miss class, it is your responsibility to find out about any announcementsor assignments you may have missed.
E-Learning	▪ It is your responsibility to check the course's eLearning website regularly (atleast once every 2-3 days) for announcements and material.
Workload	▪ The student should expect to spend 6 hours per week as an average workload.
Graded Exams	▪ Instructor should return exam papers graded to students within one week afterthe exam date.
Participation	▪ Participation in and contribution to class discussions will affect your finalgrade positively. Raise your hand if you have any questions. ▪ Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	▪ Assignments are as mentioned in the previous table. ▪ Students organize themselves in teams of (2-3) students each. Each team willgive at least one presentation regarding a selected case, delivering a fully working application. ▪ More details about phases and topics of the projects will be given later on.
Others	

Course Policies

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Rubrics and Feedbacks (If Applicable)

* **JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3	13.5	40.5
Tutorials	4	5	20
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	0	0	0
Self-Learning Activities			
Assignments/ Homework	3	5	15
Case Studies	3	1	3
Presentations	3	1	3
E-learning			
Extra Readings	3	12	36
Total Hours	120.5		
JNQF Hours	**=Total Hours/10** = 120.5/10 = **12.05**		